

CLEARDALE AGDM, AB, Canada

WMO: 712210

Lat: 56.3144N

Lon: 119.7450W

Elev: 2067

StdP: 13.63

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.0	-22.3	-35.8	0.8	-28.9	-28.7	1.2	-21.9	23.2	36.0	20.9	36.2	2.1	280	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	81.4	61.0	78.0	59.6	75.1	58.5	64.0	75.9	62.0	73.9	60.3	71.7	7.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	81.8	68.2	57.5	76.1	66.2	55.6	71.1	64.0	30.3	76.2	28.8	74.1	27.5	71.8	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	17.0	14.6	-36.4	87.1	5.5	3.6	-40.3	89.7	-43.6	91.8	-46.7	93.8	-50.7	96.4
			-36.4	67.2	5.3	3.0	-40.3	69.3	-43.4	71.1	-46.4	72.7	-50.2	74.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.1	9.8	11.8	22.3	37.7	49.9	57.0	60.7	57.8	49.3	37.1	18.1	8.2
	DBStd	22.08	16.53	14.73	13.57	8.61	7.52	5.31	4.65	5.84	7.46	8.56	13.13	15.07
	HDD50	6414	1246	1070	859	374	94	6	0	7	102	404	958	1294
	HDD65	10936	1711	1490	1324	819	468	244	148	230	472	864	1407	1759
	CDD50	980	0	0	0	5	92	217	331	250	80	5	0	0
	CDD65	27	0	0	0	0	0	5	14	8	0	0	0	0
	CDH74	881	0	0	0	2	53	156	372	242	55	1	0	0
	CDH80	158	0	0	0	0	5	22	73	51	7	0	0	0
	WSAvg	5.5	4.9	4.5	5.3	6.4	6.4	6.0	5.9	5.5	5.9	5.9	4.6	4.6
	PrecAvg	18.1	1.2	0.9	1.0	0.8	1.3	2.9	3.1	2.0	1.3	1.1	1.2	1.2
(15) Precipitation	PrecMax	25.8	2.6	1.8	2.7	1.8	3.4	6.0	5.7	4.4	3.3	3.7	3.5	5.1
	PrecMin	11.3	0.4	0.0	0.2	0.2	0.3	1.0	1.1	0.3	0.3	0.3	0.2	0.2
	PrecStd	3.9	0.6	0.5	0.6	0.4	0.8	1.5	1.1	1.1	0.7	0.7	0.7	1.0
	PrecStd	3.9	0.6	0.5	0.6	0.4	0.8	1.5	1.1	1.1	0.7	0.7	0.7	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	44.2	52.1	69.5	80.0	83.1	86.4	85.2	79.9	68.2	50.6	43.9
		MCWB	38.1	37.7	41.2	49.6	56.3	61.0	64.0	62.1	58.2	52.5	41.8	37.2
	2%	DB	42.1	41.3	47.3	61.8	74.2	78.2	81.7	80.0	74.2	59.0	44.5	40.3
		MCWB	36.9	35.2	38.5	46.3	53.7	58.5	62.2	61.7	57.4	47.1	37.9	34.8
	5%	DB	39.3	38.4	44.2	56.6	70.1	74.9	78.1	76.0	68.6	54.2	40.4	35.7
		MCWB	34.7	33.4	36.5	43.2	51.3	57.4	60.8	59.4	54.4	44.6	35.0	31.1
	10%	DB	34.4	34.1	40.9	52.2	66.2	71.5	74.7	72.5	63.7	51.0	36.2	31.5
		MCWB	30.8	29.9	34.4	40.8	49.8	56.1	59.8	58.5	52.0	42.6	32.2	28.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	38.4	42.3	50.8	58.2	64.3	67.7	65.7	61.7	53.7	42.3	38.0
		MCDB	43.9	43.4	50.5	67.7	74.4	78.7	80.3	76.9	74.8	67.3	49.7	43.3
	2%	WB	37.1	35.9	39.0	46.9	55.0	61.1	64.6	63.4	58.2	48.6	38.6	35.0
		MCDB	42.0	40.7	46.5	60.9	71.0	73.4	76.2	75.5	71.3	56.8	44.0	40.1
	5%	WB	34.7	33.5	36.8	43.7	52.8	59.1	62.7	61.3	55.7	45.6	35.2	31.5
		MCDB	38.9	38.1	43.6	54.6	66.9	71.0	74.1	73.2	67.0	52.7	39.7	35.1
	10%	WB	31.1	29.9	34.5	41.3	50.9	57.2	60.9	59.3	53.0	43.2	32.3	28.0
		MCDB	34.7	34.2	40.4	50.9	63.9	68.4	72.0	70.1	62.2	50.1	35.4	31.3
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	20.2	21.2	20.8	24.1	23.5	23.8	24.3	22.9	18.0	16.7	16.9
		MCDBR	19.6	22.7	22.5	29.2	31.5	30.2	31.5	31.6	31.7	24.9	20.3	21.8
		MCWBR	15.4	17.1	15.2	15.8	14.9	14.2	14.5	14.8	16.2	14.9	15.3	17.7
		MCDBR	19.4	21.5	21.3	27.4	28.6	25.8	27.9	27.9	30.6	23.9	19.6	21.3
	5% WB	MCWBR	15.4	16.4	14.7	15.2	14.1	12.8	14.4	14.5	16.3	15.0	15.3	17.7
(40) Clear-Sky Solar Irradiance	taub		0.251	0.254	0.287	0.320	0.346	0.348	0.350	0.346	0.306	0.284	0.240	0.219
	taud		2.209	2.324	2.286	2.376	2.398	2.409	2.430	2.436	2.555	2.565	2.405	2.237
	Ebn at Noon		209	265	280	286	283	284	280	274	273	254	223	198
	Edh at Noon		17	23	30	33	34	34	33	31	23	18	14	12
(44) All-Sky Solar Radiation	RadAvg		214	516	991	1490	1761	1901	1845	1502	1028	541	240	148
	RadStd		22	41	71	76	121	161	107	133	87	62	21	11

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page